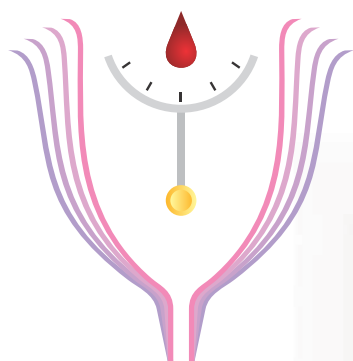




Ref. Code : 19MH1852046

Inspire Award State Level Exhibition

PICTORIAL JOURNEY OF PROJECT



POST PARTUM CUP

A novel way to estimate the obstetric
blood loss accurately and save
Maternal Life in **Golden Hour**

**SAVE MOTHER
&
SAVE MOTHER EARTH**

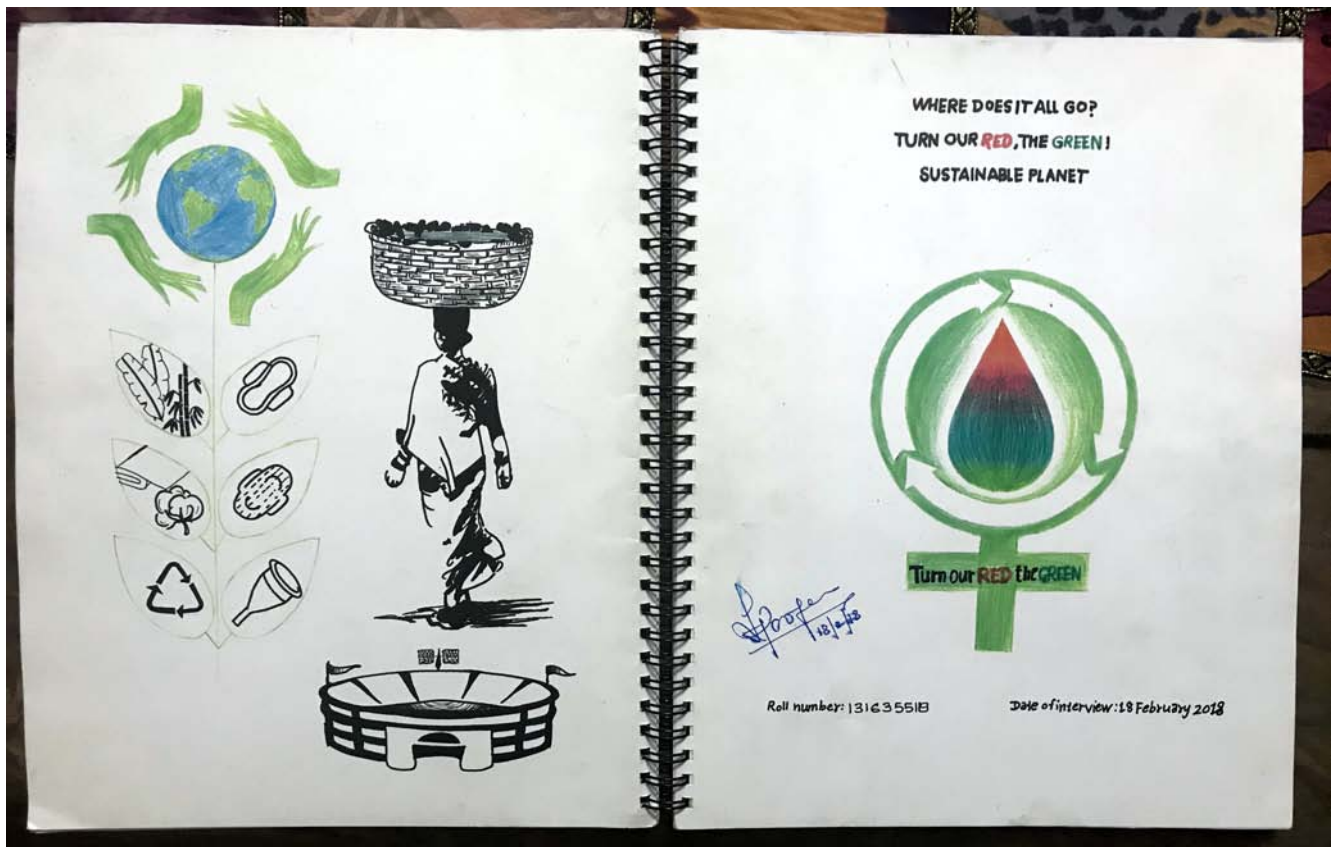
Project by : Archit Rahul Patil

Guide : Mrs. Sangita Khanna

Principal : Mr. Amit Bhatia Sir,
Kashinath Palod Public School, Jalgaon.



My journey starts with the project (“Turn our Red the Green”) for Homibhabha Balvaidyanik Competition 2017-18 honored with **Gold Medal**. I found that menstrual cup made up of medical grade silicone is a **cash free, rash free and trash free** menstrual blood collection product, is being used and is very good menstrual hygiene option. During my project, I found excellent acceptance of menstrual cup by volunteers including my mother and other three gynecologists.



'Gold Medal' & certificate received in presence of renowned scientist Dr. Anil Kakodkar



Turn our **Red** the **Green**



Statement of Necessity

Every day sanitary waste occupying nearly 22 football ground is generated all over the world. Which remains in the environment for 100's of years as it contains plastic. It also contain chemicals like dioxin which is carcinogenic. Sanitary pads are practically difficult to dispose. Scientific and ideal incinerators are not available. When sanitary pads are burns, it releases harmful gases.

Miserable Ground Reality @ Jalgaon



Interactive Workshops



Interactive Session at Nursing Students



Turn our **Red** the **Green**

School of Environmental & Earth Sciences (North Maharashtra University, Jalgaon)



Print Media

जळगावातील 'लिटल पॅडमॅन' अर्चित पार्टील

नॅपकीनच्या धुरामुळे कॅन्सर, वंध्यत्व, अंडाशयाचे आजार होण्याची शक्यता

अशा आहेत उपाययोजना

जळगाव : सध्या सर्वत्र 'पॅडमॅन' या चित्रपटाची सर्वत्र चर्चा असून मासिक 'पॅडमॅन' या विषयावर अनेक लोक खुली चर्चा करीत आहेत. त्यामुळे अशा विविध विषयांना हात घालण्याचा प्रयत्न पॅडमॅन मधून करण्यात येत आहे. याच अनुषंगाने जळगावच्या एका सारवाचीच्या विद्यालयात सैनटरी नॅपकीनच्या वाढत्या वापरामुळे होणारे अजार व प्रदूषणाचा आज्ञा पाळण्यासाठी एक प्रोजेक्ट तयार केला आहे. या प्रोजेक्टमधून त्यांचा उपाययोजना सांगितल्या आहेत.

मासिक पाळीसारखा महत्वाचा विषय आहे. आताही आपल्याकडे गुलाब पाळली नाही. मासिक पाळीदरम्यान खिचणीच असेल, स्वच्छता बसायला विषयावर बोललेले देखील जात नाही. नेमका याच विषयावर अक्षय कुमारचा 'पॅडमॅन' भाष्य करणार आहे. मात्र, यापेक्षाही दोन पाऊले पुढे जाऊन जळगावातील कॉमिनायब पलीड या शाळेतील सारवाचीच्या विद्यापीठी अर्चित पार्टील यांनी एक प्रोजेक्ट तयार केला आहे. सैनटरी नॅपकीनचा दिवसीयस वाढता वापर आणि त्याबद्दल निर्माण होणारा कचरा व प्रदूषण टाळण्यासाठी कचरावाच्य उपाययोजना या प्रोजेक्टमधून दर्शविल्या आहेत.

असा निवडला विषय

अर्चित हा शाळातील निवडलेला डॉ. राहुल पार्टील व भूलनज डॉ. अर्चना पार्टील यांचा

एक जण निर्माण करत १२० ते १५० किलो कचरा

मासिक पाळीदरम्यान मुली व महिलांकडून सैनटरी नॅपकीनचा वापर मोठ्या प्रमाणात होत आहे. पूर्वी वापरण्यात येणाऱ्या कापडाचा वापर हा कमी झाला असून प्रतिठेचा विषय हा बनला आहे. त्यामुळे एका वर्षासाठी एक महिला मासिक पाळीदरम्यान १४४ सैनटरी नॅपकीन वापरते असे एका सर्वेक्षण समोर आले आहे. म्हणजेच १३ ते ५० टक्केपर्यंत एक महिला ५,३२८ एवढे सैनटरी नॅपकीन वापरते. परिणामी सैनटरी नॅपकीनचा एक महिला आपल्या आयुष्यात १२० ते १५० किलो कचरा निर्माण करते, असेही सर्वेक्षणानुसार समोर आले आहे.

२ वर

Electronic Media

जळगावचा 'पॅड मॅन'

जळगावचा 'पॅड मॅन' या चित्रपटाची सर्वत्र चर्चा असून मासिक 'पॅडमॅन' या विषयावर अनेक लोक खुली चर्चा करीत आहेत. त्यामुळे अशा विविध विषयांना हात घालण्याचा प्रयत्न पॅडमॅन मधून करण्यात येत आहे. याच अनुषंगाने जळगावच्या एका सारवाचीच्या विद्यालयात सैनटरी नॅपकीनच्या वाढत्या वापरामुळे होणारे अजार व प्रदूषणाचा आज्ञा पाळण्यासाठी एक प्रोजेक्ट तयार केला आहे. या प्रोजेक्टमधून त्यांचा उपाययोजना सांगितल्या आहेत.

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२ वर



https://www.youtube.com/watch?v=dzwh7ou2a_E



Conclusion

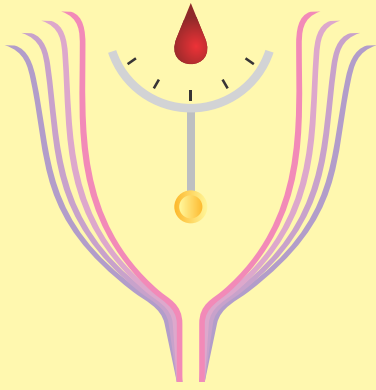
Menstrual cup (made up of medical grade silicone) is best available menstrual hygiene option.

Cash Free... Rash Free... Trash Free...

Brain Twister

The idea struck to my mind, if this menstrual cup is modified to fit the drainage tube and connected to collection bag (like a urine bag). It can be used in post delivery (postpartum) patient and will do real-time estimation of post delivery (postpartum) blood loss.





POST PARTUM CUP

A novel way to estimate the obstetric **blood loss** accurately and save **Maternal Life in Golden Hour**

How the idea struck first?

As my father is obstetrician and mother is anesthesiologist, they get distress phone calls for help in cases of postpartum hemorrhage at odd hours of night. Later I use to hear their discussion how marathon efforts were needed to put by all medicos to save the maternal life {like operative interventions-compression sutures, various balloon tamponed, stepwise de-vascularisation : even internal iliac artery ligation, sometimes obstetric hysterectomy ; when all measures failed. (I used see photos and videos of these procedures)} .



The delay or under diagnosis was responsible for landing the mothers in near-miss morbid condition which might cause maternal death.

Novel Invention :

I thought, what can be done to help obstetricians to eliminate the delay and can do real time estimation of post partum bleeding? The new method should be user friendly and acceptable to patient, nursing staff and obstetrician. **Accurate estimation of obstetric blood loss is not possible with any of prevalent methods available in clinical practice.**

Statement of Necessity

PPH (Post Partum Haemorrhage) is major cause of maternal death .

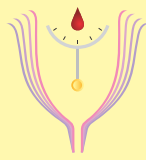
Globally one mother dies every 4 minute due to PPH.

We are still living in the Shadow of Tajmahal which reminds us the grim tragedy of death of Mughal Empress 'Mumtaz' who died during 14th child birth due to PPH.

While India's MMR (Maternal Mortality Rate) is 130. It is better than some of its neighbours, such as Pakistan (178), Bangladesh (176), Nepal (258). But it is substantially behind Sri Lanka (30), according to 2015 World Bank data, Fact Checker.in reported on June 19, 2018.

India accounts for 17% of the global burden of maternal deaths. The leading causes of such deaths in India are haemorrhage (38%).

However, India's current MMR is still higher than the sustainable development goal (SDG) target--a set of globally agreed goals that India has signed on to--of 70 deaths per 1,00,000 live births for the world by 2030.



POST PARTUM CUP



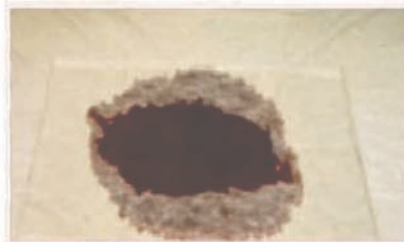
soiled sanitary towel
30ml



saturated small swab (10x10cm)
60 ml



saturated sanitary towel
100 ml



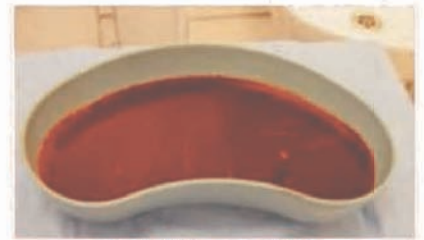
incontinence pad
250 ml



saturated swab (45x45cm)
350 ml



floor spill (100cm diameter)
500 ml



full kidney dish
500 ml



blood spilling on bed
1000 ml



blood spilling to floor
2000 ml

Brass-V Drape

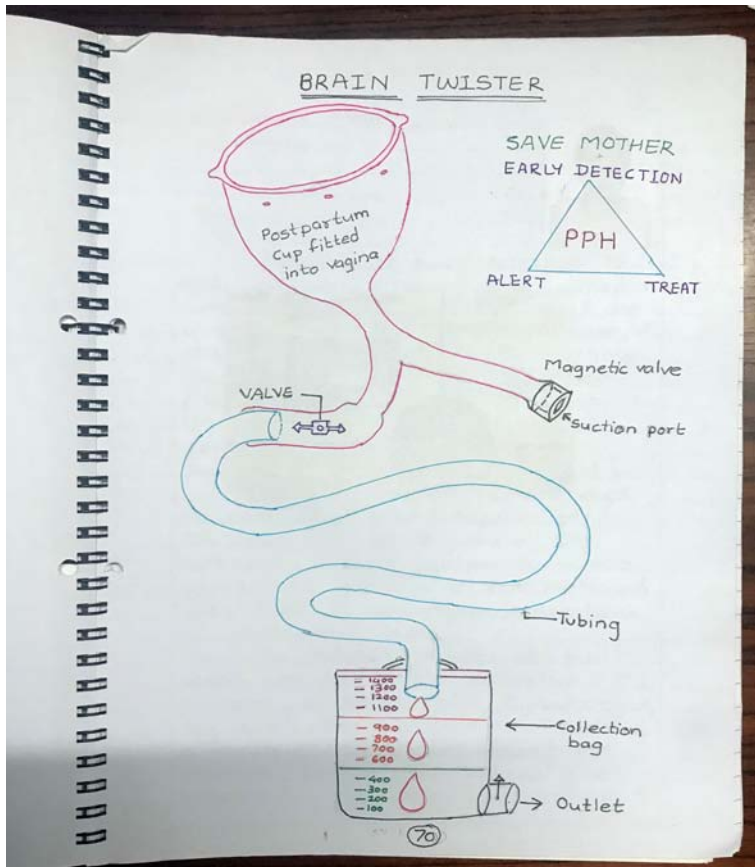


Kelly's Pad





POST PARTUM CUP



First the idea was put on the paper.



Preparing primitive working model





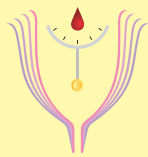
POST PARTUM CUP



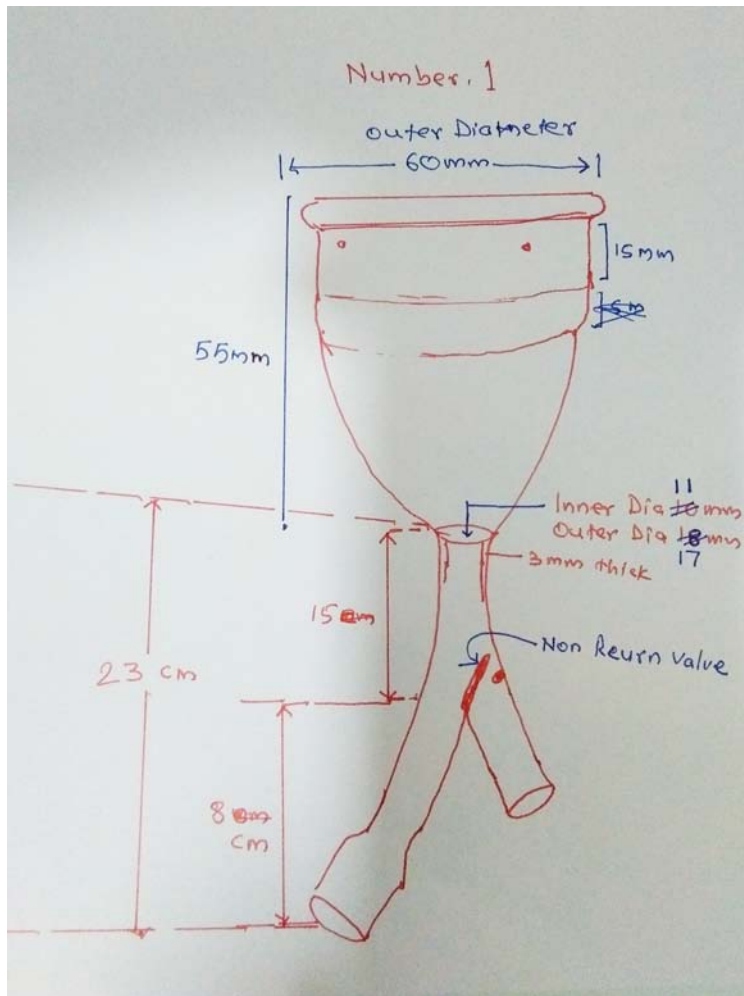
**Presenting the
Primitive working model
& Idea of PPH Cup
to Gynecologist.**



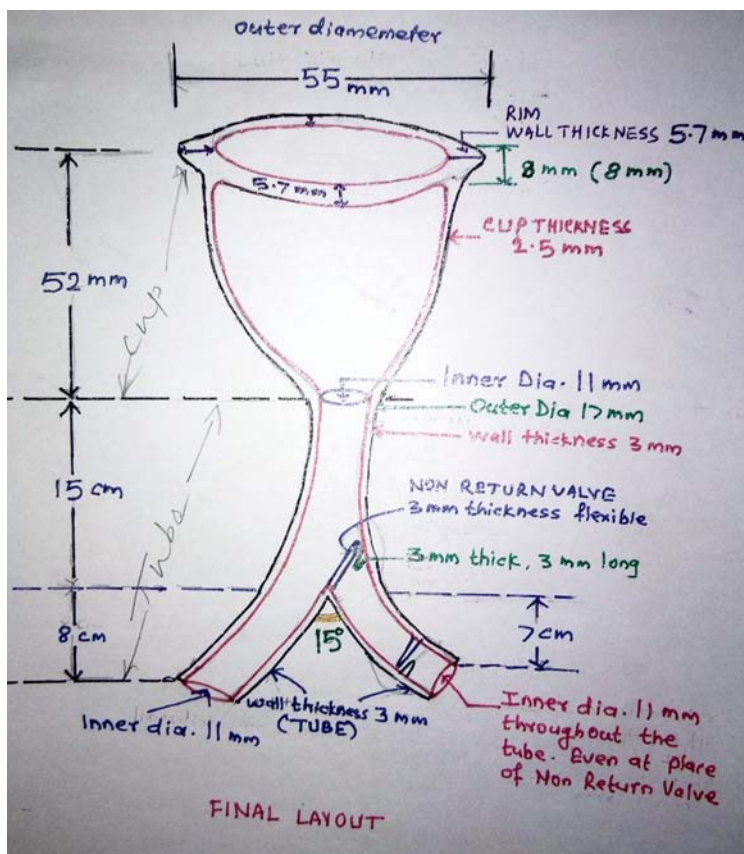
**Primitive working model
project presented at
KPPS, Jalgaon at
School Level.**



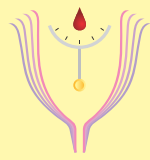
POST PARTUM CUP



Developing 2D Layout of PPH Cup



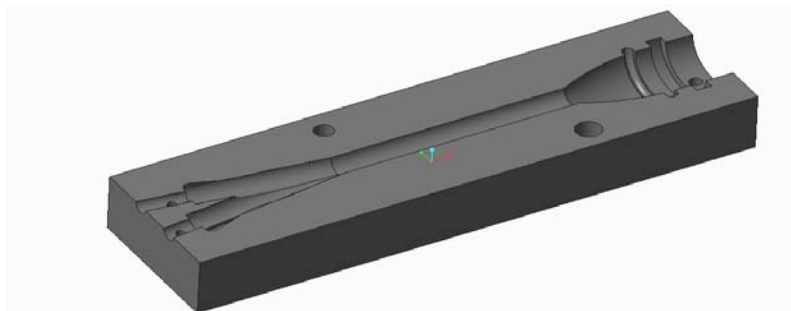
Final Layout and fixing dimensions for production of PPH Cup



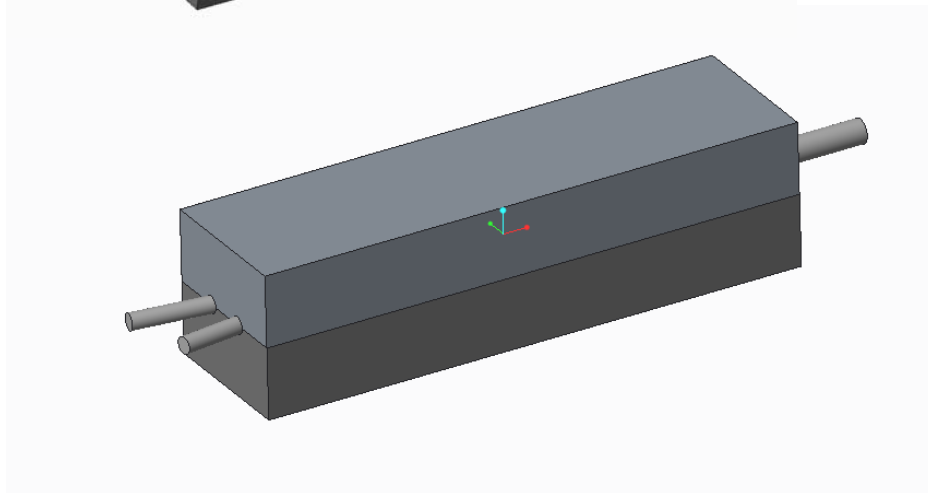
POST PARTUM CUP



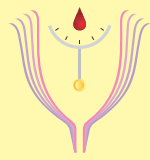
Developing 3D Model



Developing Mould



Actual Mould



POST PARTUM CUP



PPH Cup ready for clinical use.



Clinical use of
PPH Cup.



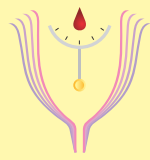
POST PARTUM CUP



Bed side monitoring of post delivery blood loss in collection bag.



Realtime & Accurate
measurement
of post delivery blood loss with
the help of PPH Cup.



POST PARTUM CUP

Conclusion

- We used the postpartum cup in the clinical practice in three maternity homes under the guidance of obstetricians and found that, it is **life saving, practical, convenient, healthier, economical, comfortable, easy to use and ecofriendly.**
- Postpartum cup is Cash free, Rash free, Trash free same as its predecessor i.e. menstrual cup
- The patients are extremely happy, comfortable and readily acceptable to the new idea as it minimized the pain, handling and excluded the feeling of wetness of postpartum blood loss.
- The nursing staff found it user friendly to handle the postpartum cup, reduced the workload of cleaning, it eliminated the use of sanitary pads and subsequent sanitary waste. Also avoided soiling of bed sheets by blood stains.
- **The obstetrician and gynecologists welcomed the idea, they confessed that first time in their clinical practice, they could see the real time estimation of post partum blood loss.** They got **early alarm** of postpartum hemorrhage and manage the patients with timely measures and saved maternal lives. Now obstetricians are over demanding the samples of postpartum cup. **Three obstetricians have incorporated this invention in their routine practice and recommending to other doctors to use it.** There is more demand for this postpartum cup from medical fraternity.
- This idea is to eliminate the under diagnosis and loss of golden hour. So timely measures are taken by the treating obstetricians and team to curb the maternal mortality.
- **It has added advantage of eliminating the sanitary biomedical waste and is eco friendly .**

LIMITATIONS / SUGGESTIONS:

Although there were no limitations, however few suggestions such as,

- 1) Different sizes should be made available.
- 2) The inner diameter of the suction port should be such that 10 mm suction tube should negotiate easily the joint of drainage end and suction end of the postpartum cup.
- 3) The collecting bag should be snugly (tight) fitting to drainage end of the cup to make it water tight.

Above mentioned suggestions can be easily implemented with modifications in the designs of same mould.



POST PARTUM CUP

IMPORTANCE OF THE WORK:

PPH is the major killer all over the world.

As concluded earlier, this idea is to eliminate the under diagnosis and loss **golden hour**. So timely measures are taken by the treating obstetricians and team to reduce the maternal mortality.

The postpartum cup works wonder in accurately estimating the postpartum obstetric blood loss.

The things should start here and society should walk the talk to save maternal life.

Every life is precious. This innovation will have impact to save many maternal lives if it is brought on the right platform and reduce the maternal mortality rate as per policy of Government of India and WHO (World Health Organization).

This novel cup reduces the sanitary waste generated in hospitals to a great extent and so is ECOFRIENDLY.

This novel idea is fit to apply for patent as no such simple yet indispensable medical tool is available in medical literature. It also has great commercial potential.

It needs to be presented on the global platform.

Brain Twister :

1. It may be possible to get blood components like pack cell volume (PCV) i.e. red blood cells and other components from the collected (obstetric) blood. If need arises, these blood components can be used for transfusion in the same patient.
2. It may be the source of Stem Cells.

Further research is needed in this regards





POST PARTUM CUP

Doctor's Feedback



नाव : _____ दि. / / 201

पता : _____
I, myself Dr. Anita sanjay Bawiskar. Working as consultant obstetrician & Gynaecologist at Krishna Hospital, Jalgaon. want to share my experience and opinion, regarding use of Post partum cup.

I am using this Postpartum cup for past 4 months for my patients. It is of great use to determine immediate blood loss after delivery as well as total blood loss in 24hrs of delivery and I can take immediate action for the management of post partum haemorrhage, which is life saving.

It also helps to determine blood loss for high risk patients like those having hypertension, twins delivery, obese patient etc. For obese patients

परत येताना हा कागद सोबत आणणे आवश्यक आहे.

to change frequent pads ~~for~~ having heavy bleeding after
is very difficult and requires manpower. Here with the help of suction machine we can remove clots very easily and thus physical handling of patient which is very painful can be definitely minimised. It also saves patients money as ~~there is~~ no requirement of changing pads for first 24 hrs. No staining of bedsheet and patients clothes too.

I really found this innovative idea of using Postpartum cup for patients - great. We can save life of patients suffering PPH which is one of the major cause of maternal mortality in our country. Thus we can help to reduce maternal mortality rate and ~~achieve~~ achieve our goal for maternal health.

DR. ANITA BAWISKAR
M.S. (Ob. & Gyn.) D.G.O.
Co-ordinator Gynaecologist
Krishna Hospital
Jalgaon-425001

Patient's Feedback

पेक्षाये नाव :- अश्विनी चेतन कोपले.
पत्ता :- चंद्रप्रभा कॉलनी, फ्लॅट नं. 18 जळगाव

Mob:-

डिलिव्हरीची तारीख :- 26/12/18

माझी डिलिव्हरी नंतर मी Post Partum cup वापर केला Post Partum cup हा उत्तम पर्याय आहे. कारण माझी पहिली डिलिव्हरी ही दुसऱ्या हॉस्पिटलमध्ये झाली तिथे मला प्रकारचा Post Partum cup वापरला नाहीता, त्या Post Partum cup मुळे वारंवार पॅड बदलावे लागले नाहीत. आणि जे वेळी बिल्ली योग्य वेळी सफाई करून काढत करता आले. आणि मला कुठल्याही प्रकारचा त्रास झाला नाही. निचमिच सुरक्षा वाटले नाही. Post Partum cup हा माझ्यासाठी परित्या डिलिव्हरी मध्ये होणाऱ्या बासा पेक्षा चांगला उपयोगी पडला.

अश्विनी चेतन कोपले

दिपाली हर्षल यादुंबे

वय :- 28

राधाकृष्ण नगर जळगाव

ता. जि. जळगाव

Mob No:- 9822375100

Delivery Date:- 13/11/2018

Postpartum cup हा माझ्या Delivery च्या दिवशी मला use करण्यात आला. त्यामुळे Pad घेणे हा त्रास तर नव्हता पण मला त्यामुळे फ्रेश वारले न माझे blood किती गेले हे पण कळले. तेंपकीन परत परत change करणे हा त्रास तर नव्हताना. Postpartum Cup हा अगदी उत्तम पर्याय आहे.

Postpartum Cup हा Best option तर आहेच पण हा एक great experience आहे.

अपली विशवायू

26/12/2019



Special Thanks

Project Guide	:	Mrs. Sangita Khanna
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